

Generalized Doob Steering via Normalized Feynman–Kac Semigroups for Branching Flows

A detailed theorem for likelihood-free reward-shifted sampling on variable-cardinality Euclidean state spaces

Prepared as a theoretical note

June 6, 2026

Abstract

This note revises and expands the previous argument. The main point is that the branching component of a Branching Flow is not a separate obstruction: after marginalization it is an atomic jump kernel on the disjoint union of Euclidean sequence spaces. Reward tilting is therefore handled by a normalized Feynman–Kac semigroup, equivalently by the time-inhomogeneous generalized Doob transform of the corresponding non-conservative semigroup. In addition to the main theorem, this version includes self-contained proofs of the supporting stochastic-analysis facts used in the argument: Bayes’ rule for changes of measure, the martingale/tower identity, the backward Kolmogorov equation, the Feynman–Kac formula and coffin-state killing, a Girsanov theorem proof in the style of Øksendal, and the product rule yielding the transformed generator $\mathcal{L}^H$. The exact theorem says that if

$$H(t, x) = \mathbb{E}_{t,x}[\exp\{\beta r(X_T)\}]$$

is positive and regular, then the generalized-Doob-transformed process has terminal law

$$q^\beta(dx) = \frac{e^{\beta r(x)} q(dx)}{\int e^{\beta r(y)} q(dy)}.$$

No terminal likelihood $q(x)$ is required. The continuous Euclidean drift changes by $a\nabla \log H$, while each split or deletion rate is multiplied by the desirability ratio $H(t, \text{post-jump})/H(t, \text{pre-jump})$.

Contents

1 Literature positioning: normalized Feynman–Kac semigroups and generalized Doob transforms	2
2 Background: the probability identities behind the construction	3
2.1 Conditional expectation and Bayes’ rule	4
2.2 Generators, martingales, and the backward equation	5
2.3 Feynman–Kac theorem and coffin-state killing	6
2.4 Girsanov theorem: the diffusion-only version of drift steering	8
2.5 Normalized Feynman–Kac semigroup, generalized Doob transform, and the product rule for $\mathcal{L}^H$	9

3	Branching Flows as Markov jump-diffusions	11
3.1	State space	11
3.2	Generator	12
3.3	Existence and non-explosion	12
4	Question i: When does H solve the backward integro-differential equation?	14
5	Question ii: Why do surviving samples have terminal law q^β?	15
6	Question iii: Why is $M_t = H(t, X_t)$ a martingale?	16
7	The exact generalized Doob theorem	17
8	Question iv: Why does the diffusion part follow from the product rule?	18
9	Approximate desirability functions	20
10	Practical likelihood-free recipe	20
11	Compact theorem statement for a paper	21

1 Literature positioning: normalized Feynman–Kac semigroups and generalized Doob transforms

For any measurable time-density ϕ and path X , write

$$A_{s,t}^\phi(X) := \int_{(s,t]} \phi_u(X_u) \text{Leb}(du), \quad 0 \leq s \leq t \leq T, \quad (1)$$

where Leb denotes Lebesgue measure on time. Equivalently, $A_{s,t}^\phi(X)$ is the total mass on $(s, t]$ of the signed time measure $\phi_u(X_u) \text{Leb}(du)$. This keeps the integrating measure explicit throughout.

The kernel used below,

$$P_{s,t}^{H,\rho} f(x) = H(s, x)^{-1} \mathbb{E}_{s,x} \left[\exp \left\{ A_{s,t}^\rho(X) \right\} H(t, X_t) f(X_t) \right], \quad (2)$$

with $H > 0$ satisfying $(\partial_t + \mathcal{L}_t)H + \rho H = 0$, is best viewed as a time-inhomogeneous generalized Doob transform of a Feynman–Kac semigroup. The compound label joining Doob and Feynman–Kac is not standard and is not used as a name in this note: the same construction appears in the literature as a Doob h -transform or h -process, a normalized Feynman–Kac semigroup, a ground-state transform, a twisted or tilted semigroup, a multiplicative-functional transform, a driven process, a Q -process, or a Schrödinger-bridge/reciprocal transform, depending on context.

The two historical ingredients are classical. Kac’s Wiener-functional formula supplies the exponential potential or killing weight, while Doob’s conditioned Brownian motion and potential-theoretic work supply the positive harmonic normalization [19, 5, 6]. For generator-transform antecedents beyond the harmonic-transform tradition, Kunita’s absolute-continuity work is a useful classical reference [21]. For a modern textbook Feynman–Kac reference, see Karatzas–Shreve [20, Ch. 5]. For the closest modern formulaic match to (2), see Chetrite–Touchette’s generalized Doob transform, including its time-dependent version and jump/diffusion generator formulas [23, Sec. IV.1, eqs. (71),(74),(78)–(80)]. The homogeneous long-time/eigenfunction version is the driven-process

transform of a tilted generator [22, eqs. (10)–(12)]. Léonard’s path-space formulation is a useful measure-theoretic umbrella: starting from a Markov law R , one tilts by endpoint factors and an exponential potential of the form $f(X_0)\exp\{-A_{0,1}^V(X)\}g(X_1)$, and the transformed law remains Markov [14].

Equivalently, if

$$Q_{s,t}^\rho g(x) = \mathbb{E}_{s,x} \left[\exp \left\{ A_{s,t}^\rho(X) \right\} g(X_t) \right] \quad (3)$$

then

$$P_{s,t}^{H,\rho} f = H(s, \cdot)^{-1} Q_{s,t}^\rho (H(t, \cdot) f). \quad (4)$$

Thus the object below is not a separate species of Markov transform; it is the conservative Markov kernel obtained by normalizing the non-conservative Feynman–Kac semigroup with a positive space-time harmonic function.

Literature name	Usual form	Relation to this note
Doob h -transform / h -process	Semigroup or generator conjugation by h	The H -normalization in (2); historically rooted in Doob’s conditioned Brownian motion [5, 6].
Feynman–Kac semigroup	$Q_{s,t}^\rho f = \mathbb{E}[e^{A_{s,t}^\rho(X)} f(X_t)]$	The non-conservative exponential path weighting before normalization; see Kac and standard stochastic-calculus treatments [19, 20].
Generalized Doob transform / driven process	$h^{-1}Lh - h^{-1}(Lh)$, with time-dependent variants	The closest exact modern terminology for (2); Chetrite–Touchette also give jump-rate and diffusion-drift formulas [22, 23].
Multiplicative-functional transform	$f(X_0)e^{-A_{0,1}^V(X)}g(X_1)R$	The abstract path-law version with endpoint factors and an interior potential [14].
Schrödinger bridge / reciprocal transform	Endpoint-biased bridges and mixtures	Conceptually adjacent when the weighting is concentrated in endpoint factors [25, 26].
Q -process / quasi-stationary transform	Survival-conditioned killed semigroup normalized by a positive eigenfunction	The absorbing/killed-process counterpart of the same normalization principle [27, 28].

This is the terminology used in the rest of the note: “normalized Feynman–Kac semigroup” emphasizes the path-weighted non-conservative semigroup, while “generalized Doob transform” emphasizes the conservative Markov normalization by H . The citations above connect the same object to the standard names used in probability, PDE/spectral theory, large deviations, Schrödinger bridges, and conditioned survival.

2 Background: the probability identities behind the construction

This section collects the exact stochastic-analysis facts used later. The style is deliberately close to Øksendal: state a useful identity, prove it, and then immediately interpret it for the steering construction.

2.1 Conditional expectation and Bayes' rule

Recall that if X is integrable and $\mathcal{G} \subset \mathcal{F}$ is a sub- σ -algebra, then $\mathbb{E}[X \mid \mathcal{G}]$ is the essentially unique $\mathcal{G}$ -measurable random variable satisfying

$$\int_G \mathbb{E}[X \mid \mathcal{G}] d\mathbb{P} = \int_G X d\mathbb{P}, \quad G \in \mathcal{G}. \quad (5)$$

The tower property says that if $\mathcal{G} \subset \mathcal{H} \subset \mathcal{F}$, then

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{G}]. \quad (6)$$

Theorem 2.1 (Bayes formula for a change of measure). *Let $\mathbb{Q}$ and $\mathbb{P}$ be probability measures on $(\Omega, \mathcal{F})$ with*

$$d\mathbb{Q} = L d\mathbb{P}, \quad L \geq 0, \quad \mathbb{E}_{\mathbb{P}}[L] = 1.$$

Let $\mathcal{G} \subset \mathcal{F}$ and let X be $\mathbb{Q}$ -integrable. Then, on the set where $\mathbb{E}_{\mathbb{P}}[L \mid \mathcal{G}] > 0$,

$$\boxed{\mathbb{E}_{\mathbb{Q}}[X \mid \mathcal{G}] = \frac{\mathbb{E}_{\mathbb{P}}[LX \mid \mathcal{G}]}{\mathbb{E}_{\mathbb{P}}[L \mid \mathcal{G}]} .} \quad (7)$$

If $d\mathbb{Q}/d\mathbb{P}$ restricted to $\mathcal{G}$ is written as $L_{\mathcal{G}} = \mathbb{E}_{\mathbb{P}}[L \mid \mathcal{G}]$, this is the usual conditional Bayes rule.

Proof. Put

$$Y = \frac{\mathbb{E}_{\mathbb{P}}[LX \mid \mathcal{G}]}{\mathbb{E}_{\mathbb{P}}[L \mid \mathcal{G}]}$$

where the denominator is positive. The random variable Y is $\mathcal{G}$ -measurable. For any $G \in \mathcal{G}$,

$$\begin{aligned} \int_G Y d\mathbb{Q} &= \int_G Y L d\mathbb{P} = \int_G Y \mathbb{E}_{\mathbb{P}}[L \mid \mathcal{G}] d\mathbb{P} \\ &= \int_G \mathbb{E}_{\mathbb{P}}[LX \mid \mathcal{G}] d\mathbb{P} = \int_G LX d\mathbb{P} = \int_G X d\mathbb{Q}. \end{aligned}$$

In the second equality we used that $Y \mathbf{1}_G$ is $\mathcal{G}$ -measurable. By the defining property (5), $Y = \mathbb{E}_{\mathbb{Q}}[X \mid \mathcal{G}]$. $\square$

Corollary 2.2 (Event Bayes rule and binary selectors). *If $A, B \in \mathcal{F}$ and $\mathbb{P}(B) > 0$, then*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}. \quad (8)$$

In particular, let $Y \sim q$ on a measurable space $(\mathbf{X}, \mathcal{B})$ and let $S : \Omega \rightarrow \{0, 1\}$ be a binary random variable. If

$$\mathbb{P}(S = 1 \mid Y = y) = \alpha(y), \quad 0 \leq \alpha \leq 1,$$

then, whenever the denominator is positive,

$$\boxed{\mathbb{P}(Y \in A \mid S = 1) = \frac{\int_A \alpha(y) q(dy)}{\int_{\mathbf{X}} \alpha(y) q(dy)} .} \quad (9)$$

The later “survival” language is just the special case in which S is interpreted as the indicator that a sample is retained rather than rejected or killed.

Proof. The event formula follows from

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad \mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}.$$

For (9), compute

$$\begin{aligned} \mathbb{P}(Y \in A \mid S = 1) &= \frac{\mathbb{P}(Y \in A, S = 1)}{\mathbb{P}(S = 1)} \\ &= \frac{\mathbb{E}[\mathbf{1}_A(Y)\mathbb{P}(S = 1 \mid Y)]}{\mathbb{E}[\mathbb{P}(S = 1 \mid Y)]} \\ &= \frac{\int_A \alpha(y)q(\mathrm{d}y)}{\int_X \alpha(y)q(\mathrm{d}y)}. \end{aligned}$$

□

Corollary 2.3 (Reward tilt as Bayes rule). *Let $w : X \rightarrow (0, \infty)$ be integrable and define*

$$q^w(\mathrm{d}y) = \frac{w(y)q(\mathrm{d}y)}{\int_X w(z)q(\mathrm{d}z)}. \quad (10)$$

If w is bounded and $M \geq \|w\|_\infty$, then q^w is the conditional law of Y given a binary acceptance variable S satisfying

$$\mathbb{P}(S = 1 \mid Y = y) = \frac{w(y)}{M}.$$

When S is interpreted as a survival indicator, this is exactly terminal survival conditioning. In particular, for $w = e^{\beta r}$ this is the reward-shifted law q^β .

Proof. Apply Corollary 2.2 with $\alpha(y) = w(y)/M$. The constant M cancels. □

2.2 Generators, martingales, and the backward equation

Let $X = (X_t)_{0 \leq t \leq T}$ be a time-inhomogeneous Markov process with natural filtration $\mathcal{F}_t = \sigma(X_s : s \leq t)$ and transition operators

$$P_{s,t}f(x) = \mathbb{E}_{s,x}[f(X_t)], \quad 0 \leq s \leq t \leq T.$$

On a suitable test class $\mathcal{D}$, its generator is

$$\mathcal{L}_t f(x) = \lim_{h \downarrow 0} \frac{\mathbb{E}_{t,x}[f(X_{t+h})] - f(x)}{h}. \quad (11)$$

Equivalently, for $f \in \mathcal{D}$,

$$f(X_t) - f(X_0) - \int_0^t \mathcal{L}_s f(X_s) \mathrm{d}s \quad (12)$$

is a martingale. This martingale-problem view is the correct one for Branching Flows, because the split/delete mechanism is a jump kernel rather than a density.

Theorem 2.4 (Backward Kolmogorov equation). *Let g be bounded measurable and define*

$$H(t, x) = \mathbb{E}_{t,x}[g(X_T)]. \quad (13)$$

Then H satisfies the dynamic programming identity

$$H(s, x) = \mathbb{E}_{s,x}[H(t, X_t)], \quad s \leq t \leq T. \quad (14)$$

This is the mild backward equation. If H is C^1 in time, lies in the domain of $\mathcal{L}_t$ in space, and the following limits are locally uniform, then H solves the classical backward equation

$$\boxed{\partial_t H(t, x) + \mathcal{L}_t H(t, \cdot)(x) = 0, \quad H(T, x) = g(x).} \quad (15)$$

Conversely, if a bounded regular function u solves (15) with terminal value g , then $u(t, x) = H(t, x)$.

Proof. The dynamic programming identity is the Markov property and the tower property:

$$H(s, x) = \mathbb{E}_{s,x}[g(X_T)] = \mathbb{E}_{s,x}[\mathbb{E}[g(X_T) \mid \mathcal{F}_t]] = \mathbb{E}_{s,x}[H(t, X_t)].$$

Now put $t = s + h$ in (14):

$$H(s, x) = \mathbb{E}_{s,x}[H(s + h, X_{s+h})].$$

Subtract $H(s + h, x)$ and divide by h :

$$\frac{H(s + h, x) - H(s, x)}{h} = -\frac{\mathbb{E}_{s,x}[H(s + h, X_{s+h})] - H(s + h, x)}{h}.$$

As $h \downarrow 0$, the left side tends to $\partial_s H(s, x)$ and the right side tends to $-\mathcal{L}_s H(s, \cdot)(x)$, giving (15).

For the converse, apply the time-inhomogeneous Ito-Dynkin formula, or equivalently the martingale problem, to $u(t, X_t)$:

$$u(T, X_T) - u(t, x) - \int_t^T (\partial_s + \mathcal{L}_s)u(s, X_s) \, ds$$

is a martingale. Since $(\partial_s + \mathcal{L}_s)u = 0$ and $u(T, \cdot) = g$, taking $\mathbb{E}_{t,x}$ gives $u(t, x) = \mathbb{E}_{t,x}[g(X_T)]$. $\square$

2.3 Feynman–Kac theorem and coffin-state killing

The exponential path weighting in this subsection is the standard Feynman–Kac construction, going back to Kac [19]; the diffusion/PDE version is treated in standard graduate form by Karatzas–Shreve [20, Ch. 5].

Theorem 2.5 (Feynman–Kac formula, terminal-time form). *Let $c : [0, T] \times \mathbf{S} \rightarrow \mathbb{R}$ be bounded below and measurable, and let g be bounded measurable. Define*

$$H(t, x) = \mathbb{E}_{t,x} \left[\exp \left\{ -\mathbf{A}_{t,T}^c(X) \right\} g(X_T) \right]. \quad (16)$$

Then H satisfies the mild identity

$$H(s, x) = \mathbb{E}_{s,x} \left[\exp \left\{ -\mathbf{A}_{s,t}^c(X) \right\} H(t, X_t) \right]. \quad (17)$$

If H is regular, then it solves

$$\boxed{\partial_t H(t, x) + \mathcal{L}_t H(t, \cdot)(x) - c(t, x)H(t, x) = 0, \quad H(T, x) = g(x).} \quad (18)$$

Conversely, any bounded regular solution of (18) equals (16).

Proof. The mild identity follows by splitting the exponential at time t and using the Markov property:

$$\begin{aligned} H(s, x) &= \mathbb{E}_{s,x} \left[e^{-A_{s,T}^c(X)} g(X_T) \right] \\ &= \mathbb{E}_{s,x} \left[e^{-A_{s,t}^c(X)} \mathbb{E} \left[e^{-A_{t,T}^c(X)} g(X_T) \mid \mathcal{F}_t \right] \right] \\ &= \mathbb{E}_{s,x} \left[e^{-A_{s,t}^c(X)} H(t, X_t) \right]. \end{aligned}$$

Expanding (17) over $[s, s+h]$ gives

$$0 = \frac{\mathbb{E}_{s,x}[H(s+h, X_{s+h})] - H(s+h, x)}{h} + \frac{H(s+h, x) - H(s, x)}{h} - c(s, x)H(s, x) + o(1),$$

which yields (18).

For uniqueness, let u be a bounded regular solution of (18). Fix $t \leq r \leq T$ and define

$$Z_r = \exp \left\{ -A_{t,r}^c(X) \right\}.$$

Since Z has finite variation, its cross-variation with $u(r, X_r)$ is zero. The product rule gives

$$\begin{aligned} d(Z_r u(r, X_r)) &= Z_r du(r, X_r) + u(r, X_r) dZ_r \\ &= Z_r [(\partial_r + \mathcal{L}_r)u(r, X_r) dr + dM_r] - Z_r c(r, X_r)u(r, X_r) dr \\ &= Z_r [(\partial_r + \mathcal{L}_r - c)u](r, X_r) dr + Z_r dM_r = Z_r dM_r, \end{aligned}$$

where M is the local martingale term from the martingale problem. Boundedness/localization makes the product a true martingale. Therefore

$$u(t, x) = \mathbb{E}_{t,x} [Z_T u(T, X_T)] = \mathbb{E}_{t,x} \left[e^{-A_{t,T}^c(X)} g(X_T) \right].$$

□

Proposition 2.6 (Coffin-state killing). *Assume $c \geq 0$. Let E be an independent exponential random variable with mean one and set*

$$\zeta = \inf \left\{ t \geq 0 : A_{0,t}^c(X) \geq E \right\}. \quad (19)$$

Define the killed process by $\tilde{X}_t = X_t$ for $t < \zeta$ and $\tilde{X}_t = \partial$ for $t \geq \zeta$, with $f(\partial) = 0$. Then

$$\mathbb{E}_{0,x}[f(\tilde{X}_t)] = \mathbb{E}_{0,x} \left[f(X_t) \exp \left\{ -A_{0,t}^c(X) \right\} \right]. \quad (20)$$

Consequently, if c is time-homogeneous and the base generator is $\mathcal{L}$, the killed generator is $\mathcal{L} - c$.

Proof. Condition on the path of X . Since E is exponential,

$$\mathbb{P}(\zeta > t \mid X) = \mathbb{P} \left(E > A_{0,t}^c(X) \mid X \right) = \exp \left\{ -A_{0,t}^c(X) \right\}.$$

Therefore

$$\mathbb{E}[f(\tilde{X}_t)] = \mathbb{E}[f(X_t) \mathbf{1}_{\{\zeta > t\}}] = \mathbb{E} [f(X_t) \mathbb{P}(\zeta > t \mid X)],$$

which is (20). The generator computation follows by differentiating at $t = 0$ and applying the Feynman–Kac formula:

$$\lim_{t \downarrow 0} \frac{\mathbb{E}_x[f(\tilde{X}_t)] - f(x)}{t} = \partial_t \mathbb{E}_x \left[e^{-A_{0,t}^c(X)} f(X_t) \right] \Big|_{t=0} = \mathcal{L}f(x) - c(x)f(x).$$

□

This is the rigorous version of the “coffin” intuition: reward weighting can be represented as survival weighting. If $w = e^{\beta r}$ is a terminal reward, one may realize it by terminal survival as in Corollary 2.3; if the reward is accumulated along the path, one realizes it by killing with rate $c = -\rho$ when $\rho \leq 0$, or by subtracting a constant from a bounded reward rate so that the killing rate is nonnegative.

2.4 Girsanov theorem: the diffusion-only version of drift steering

The generalized Doob transform below handles jumps and diffusions simultaneously. For intuition, the continuous Euclidean drift shift is exactly the same drift shift that appears in Girsanov’s theorem; related control and variational interpretations of driven processes are developed by Chetrite–Touchette [24].

Theorem 2.7 (Girsanov theorem, bounded case). *Let B be an m -dimensional Brownian motion under $\mathbb{P}$ with filtration $(\mathcal{F}_t)_{0 \leq t \leq T}$. Let η_t be bounded and progressively measurable, and define*

$$Z_t = \exp \left\{ \int_0^t \eta_s \cdot dB_s - \frac{1}{2} \int_0^t |\eta_s|^2 ds \right\}. \quad (21)$$

Let $\mathbb{Q}$ be given by $d\mathbb{Q} = Z_T d\mathbb{P}$ on $\mathcal{F}_T$. Then

$$\hat{B}_t = B_t - \int_0^t \eta_s ds, \quad 0 \leq t \leq T, \quad (22)$$

is Brownian motion under $\mathbb{Q}$.

Proof. By Ito’s formula for stochastic exponentials,

$$dZ_t = Z_t \eta_t \cdot dB_t. \quad (23)$$

Boundedness of η implies that Z is a true martingale and $\mathbb{E}[Z_T] = 1$.

We prove that each $\hat{B}^i$ is a $\mathbb{Q}$ -martingale. By Theorem 2.1, it is enough to prove that $Z_t \hat{B}_t^i$ is a $\mathbb{P}$ -martingale. Using the product rule,

$$\begin{aligned} d(Z_t \hat{B}_t^i) &= Z_t d\hat{B}_t^i + \hat{B}_t^i dZ_t + dZ_t d\hat{B}_t^i \\ &= Z_t (dB_t^i - \eta_t^i dt) + \hat{B}_t^i Z_t \eta_t \cdot dB_t + Z_t \eta_t^i dt \\ &= Z_t dB_t^i + \hat{B}_t^i Z_t \eta_t \cdot dB_t, \end{aligned}$$

which is a stochastic integral and hence a local $\mathbb{P}$ -martingale; bounded localization gives the true martingale claim. Therefore, for $s \leq t$,

$$\mathbb{E}_{\mathbb{Q}}[\hat{B}_t^i | \mathcal{F}_s] = \frac{\mathbb{E}_{\mathbb{P}}[Z_t \hat{B}_t^i | \mathcal{F}_s]}{\mathbb{E}_{\mathbb{P}}[Z_t | \mathcal{F}_s]} = \frac{Z_s \hat{B}_s^i}{Z_s} = \hat{B}_s^i.$$

Similarly, or pathwise, the quadratic covariations are unchanged by subtracting finite-variation processes:

$$\langle \hat{B}^i, \hat{B}^j \rangle_t = \langle B^i, B^j \rangle_t = \delta_{ij} t.$$

By the Levy characterization of Brownian motion, $\hat{B}$ is m -dimensional Brownian motion under $\mathbb{Q}$. $\square$

Corollary 2.8 (Drift shift). *If under $\mathbb{P}$*

$$dX_t = b_t(X_t) dt + \sigma_t(X_t) dB_t, \quad (24)$$

then under $\mathbb{Q}$ from Theorem 2.7, using $dB_t = d\hat{B}_t + \eta_t dt$,

$$dX_t = (b_t(X_t) + \sigma_t(X_t)\eta_t) dt + \sigma_t(X_t) d\hat{B}_t. \quad (25)$$

In the generalized Doob transform below, $\eta_t = \sigma_t(X_t)^\top \nabla \log H(t, X_t)$, so the drift becomes $b + a \nabla \log H$ with $a = \sigma \sigma^\top$.

2.5 Normalized Feynman–Kac semigroup, generalized Doob transform, and the product rule for $\mathcal{L}^H$

Let $H > 0$ be a regular positive space-time function and let ρ be a bounded reward density with respect to Lebesgue time. Define the normalized Feynman–Kac kernel

$$P_{s,t}^{H,\rho} f(x) = \frac{1}{H(s,x)} \mathbb{E}_{s,x} \left[\exp \left\{ A_{s,t}^\rho(X) \right\} H(t, X_t) f(X_t) \right]. \quad (26)$$

When H solves

$$(\partial_t + \mathcal{L}_t)H + \rho H = 0, \quad (27)$$

we have $P_{s,t}^{H,\rho} \mathbf{1} = 1$, so (26) is a Markov transition kernel. This is the normalized Feynman–Kac semigroup, equivalently the generalized Doob transform of the tilted non-conservative semigroup; compare Chetrite–Touchette’s time-dependent transform [23, Sec. IV.1, eqs. (71),(74)] and Léonard’s path-space transform with endpoint factors and potentials [14].

Theorem 2.9 (Generator of the normalized Feynman–Kac semigroup). *Assume that H is positive and regular and solves (27). For a time-independent test function f , the transformed generator is*

$$\boxed{\mathcal{L}_t^H f(x) = \frac{1}{H(t,x)} \mathcal{L}_t(H_t f)(x) - f(x) \frac{1}{H(t,x)} \mathcal{L}_t H_t(x)}, \quad (28)$$

where $H_t(x) = H(t, x)$. Equivalently, before using (27),

$$\mathcal{L}_t^H f = \frac{1}{H_t} [(\partial_t + \mathcal{L}_t + \rho_t)(H_t f) - f(\partial_t + \mathcal{L}_t + \rho_t)H_t]. \quad (29)$$

For diffusions this matches the transformed generator and drift formulas in Chetrite–Touchette [23, eqs. (79)–(80)]; for jump processes it matches the corresponding multiplicative change of jump rates [23, eq. (78)]. In the time-homogeneous eigenfunction setting it reduces to the usual normalized tilted generator [22, eqs. (10)–(12)].

Proof. Starting from (26),

$$\begin{aligned} \mathcal{L}_t^H f(x) &= \lim_{h \downarrow 0} \frac{P_{t,t+h}^{H,\rho} f(x) - f(x)}{h} \\ &= \frac{1}{H_t(x)} \lim_{h \downarrow 0} \frac{\mathbb{E}_{t,x}[e^{A_{t,t+h}^\rho(X)} H_{t+h}(X_{t+h}) f(X_{t+h})] - H_t(x) f(x)}{h} \\ &\quad - f(x) \frac{1}{H_t(x)} \lim_{h \downarrow 0} \frac{\mathbb{E}_{t,x}[e^{A_{t,t+h}^\rho(X)} H_{t+h}(X_{t+h})] - H_t(x)}{h}. \end{aligned}$$

The two limits are respectively $(\partial_t + \mathcal{L}_t + \rho_t)(H_t f)(x)$ and $(\partial_t + \mathcal{L}_t + \rho_t)H_t(x)$, giving (29). Since f has no explicit time-dependence, the ∂_t and ρ terms cancel between the two brackets, yielding (28). $\square$

Now suppose $\mathcal{L}_t = \mathcal{A}_t + \mathcal{J}_t$, where on a fixed Euclidean stratum

$$\mathcal{A}_t f = b_t \cdot \nabla f + \frac{1}{2} a_t : \nabla^2 f, \quad (30)$$

while the jump part is

$$\mathcal{J}_t f(x) = \int [f(y) - f(x)] Q_t(\mathrm{d}y; x). \quad (31)$$

Lemma 2.10 (Diffusion product rule). *For smooth H and f ,*

$$\mathcal{A}_t(Hf) = H\mathcal{A}_t f + f\mathcal{A}_t H + (a_t \nabla H) \cdot \nabla f. \quad (32)$$

Therefore the diffusion part of $\mathcal{L}^H$ is

$$\mathcal{A}_t^H f = (b_t + a_t \nabla \log H) \cdot \nabla f + \frac{1}{2} a_t : \nabla^2 f. \quad (33)$$

Proof. Use

$$\begin{aligned} \nabla(Hf) &= H\nabla f + f\nabla H, \\ \nabla^2(Hf) &= H\nabla^2 f + f\nabla^2 H + \nabla H \otimes \nabla f + \nabla f \otimes \nabla H. \end{aligned}$$

Since a_t is symmetric,

$$\frac{1}{2} a_t : (\nabla H \otimes \nabla f + \nabla f \otimes \nabla H) = (a_t \nabla H) \cdot \nabla f.$$

Substituting these identities into (30) gives (32). Then

$$\begin{aligned} H^{-1} \mathcal{A}_t(Hf) - f H^{-1} \mathcal{A}_t H &= \mathcal{A}_t f + H^{-1} (a_t \nabla H) \cdot \nabla f \\ &= (b_t + a_t \nabla \log H) \cdot \nabla f + \frac{1}{2} a_t : \nabla^2 f. \end{aligned}$$

□

Lemma 2.11 (Jump product rule). *For the jump generator (31),*

$$H(x)^{-1} \mathcal{J}_t(Hf)(x) - f(x) H(x)^{-1} \mathcal{J}_t H(x) = \int \frac{H(y)}{H(x)} [f(y) - f(x)] Q_t(\mathrm{d}y; x). \quad (34)$$

Thus the Doob transform multiplies the jump kernel by $H(y)/H(x)$.

Proof. Expand the left hand side:

$$\begin{aligned} & \frac{1}{H(x)} \int [H(y)f(y) - H(x)f(x)] Q_t(\mathrm{d}y; x) \\ & - \frac{f(x)}{H(x)} \int [H(y) - H(x)] Q_t(\mathrm{d}y; x) \\ & = \int \left[\frac{H(y)}{H(x)} f(y) - f(x) - \frac{H(y)}{H(x)} f(x) + f(x) \right] Q_t(\mathrm{d}y; x) \\ & = \int \frac{H(y)}{H(x)} [f(y) - f(x)] Q_t(\mathrm{d}y; x). \end{aligned}$$

□

Lemmas 2.10 and 2.11 are the local calculation behind the main theorem. They are also the Branching Flow analogue of Øksendal's Doob h -transform exercise for Brownian motion, where the generator becomes $f \mapsto \Delta(hf)/(2h)$ when h is harmonic.

3 Branching Flows as Markov jump-diffusions

3.1 State space

For the cleanest theorem, impose a finite maximum cardinality $N < \infty$ and define

$$S_N = \bigsqcup_{n=1}^N E_n, \quad E_n = (\mathbb{R}^d)^n. \quad (35)$$

The uncapped case can be treated by a Lyapunov non-explosion condition; see Remark 3.5.

Topology and measurable structure. The symbol $\bigsqcup$ is used in the topological sense: S_N is the coproduct, or topological sum, of the strata $E_1, \dots, E_N$. Thus a set $U \subset S_N$ is open if and only if $U \cap E_n$ is open in E_n for each n . Equivalently, each stratum is a clopen component, so the cardinality map $\ell(x) = n$ for $x \in E_n$ is continuous and measurable. This is the natural topology for ordered variable-length lists, which is precisely the form used in Branching Flows: at time t the state is an ordered list $X_t = (X_t^{(1)}, \dots, X_t^{(L_t)})$ of variable length. Standard references in descriptive set theory note that Polish spaces are closed under countable topological sums; see, for example, Kechris [13, Section 3] and Srivastava [18, Chapter 2]. This matters probabilistically because Markov kernels, regular conditional probabilities, and martingale-problem formulations are cleanest on standard Borel, and in particular Polish, state spaces; see Parthasarathy [16] and Ethier–Kurtz [7].

Lemma 3.1 (The finite-list state space is Polish). *If E is Polish, then*

$$S_\infty = \bigsqcup_{n \geq 1} E^n$$

with the topological-sum topology is Polish. Hence every capped space S_N is Polish. Its Borel σ -algebra is

$$\mathcal{B}(S_\infty) = \{A \subset S_\infty : A \cap E^n \in \mathcal{B}(E^n) \text{ for every } n \geq 1\}. \quad (36)$$

Moreover, if $E = \mathbb{R}^d$, the split and deletion maps in (37)–(38) are continuous maps between strata, hence Borel maps on the disjoint union.

Proof. Because E is Polish, each finite product E^n is Polish. Choose a complete separable metric d_n on E^n and replace it by the bounded complete metric $\bar{d}_n = d_n \wedge 1$. On the disjoint union write points as pairs (n, x) with $x \in E^n$, and define

$$d((n, x), (m, y)) = \begin{cases} \bar{d}_n(x, y), & n = m, \\ 1, & n \neq m. \end{cases}$$

This metric generates the topological-sum topology: balls of radius less than one stay inside a single stratum, and inside that stratum the metric is exactly $\bar{d}_n$. It is separable because a countable union of countable dense subsets is countable and dense. It is complete because every Cauchy sequence is eventually contained in one stratum, where it converges by completeness of $\bar{d}_n$. The Borel identity (36) follows from the fact that the strata are countably many clopen components. Finally, for $E = \mathbb{R}^d$, duplication and coordinate deletion are continuous Euclidean maps $E^n \rightarrow E^{n+1}$ and $E^n \rightarrow E^{n-1}$, respectively, and therefore Borel as maps into the coproduct. $\square$

Remark 3.2 (Ordered lists versus point configurations). The state space above is the ordered-list space. This matches the Branching Flow construction, where positions in a sequence matter and

split/delete operators act on the i th element. If a problem is genuinely unordered, one can instead quotient E^n by the finite permutation group $\mathfrak{S}_n$, or represent an unordered state as a finite counting measure $\sum_i \delta_{x_i}$. Those alternatives are standard in configuration-space and point-process theory; see Daley–Vere-Jones [11, Appendix A2] and Kallenberg [12, Chapter 4]. The present theorem deliberately avoids quotient singularities by working with the ordered state space used by the sampler.

For $x = (x_1, \dots, x_n) \in E_n$, define

$$\text{split}_i x = (x_1, \dots, x_i, x_i, x_{i+1}, \dots, x_n) \in E_{n+1}, \quad (37)$$

$$\text{del}_i x = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \in E_{n-1}. \quad (38)$$

At $n = N$, split rates are set to zero. At $n = 1$, deletion rates are set to zero if the empty sequence is disallowed.

3.2 Generator

For a function $f : \mathbb{S}_N \rightarrow \mathbb{R}$, write $f_n = f|_{E_n}$. A Euclidean Branching Flow generator has the form

$$\begin{aligned} (\mathcal{L}_t f)_n(x) &= (\mathcal{A}_t^n f_n)(x) + \sum_{i=1}^n \lambda_{t,i}^{+,n}(x) [f_{n+1}(\text{split}_i x) - f_n(x)] \\ &\quad + \sum_{i=1}^n \lambda_{t,i}^{-,n}(x) [f_{n-1}(\text{del}_i x) - f_n(x)], \end{aligned} \quad (39)$$

where

$$(\mathcal{A}_t^n f_n)(x) = b_t^n(x) \cdot \nabla_n f_n(x) + \frac{1}{2} \text{Tr} \left(a_t^n(x) \nabla_n^2 f_n(x) \right), \quad a_t^n = \sigma_t^n (\sigma_t^n)^\top. \quad (40)$$

For Branching Flows, the learned marginal split and deletion rates have the form

$$\lambda_{t,i}^{+,n}(x) = h_{\text{split}}(t) R_{\theta,t,i}(x), \quad \lambda_{t,i}^{-,n}(x) = h_{\text{del}}(t) \rho_{\theta,t,i}(x). \quad (41)$$

The split/delete component can also be written as an atomic jump kernel

$$Q_t(dy; x) = \sum_{i=1}^n \lambda_{t,i}^{+,n}(x) \delta_{\text{split}_i x}(dy) + \sum_{i=1}^n \lambda_{t,i}^{-,n}(x) \delta_{\text{del}_i x}(dy). \quad (42)$$

Then the jump part is

$$\int_{\mathbb{S}_N} [f(y) - f(x)] Q_t(dy; x). \quad (43)$$

This kernel notation is important: no density, Jacobian, or likelihood appears. The split map is singular as a map into E_{n+1} because it duplicates an element exactly, but it is a perfectly valid atom of a jump kernel.

3.3 Existence and non-explosion

Lemma 3.3 (A sufficient non-explosion condition). *Assume that the split/delete rates are nonnegative Borel functions, with the boundary rates set to zero as above. Assume that on each stratum E_n , b_t^n and σ_t^n are locally Lipschitz in x and have at most linear growth, uniformly on compact time intervals. Assume also that the total jump rate*

$$\Lambda_t(x) = \sum_i \lambda_{t,i}^{+,n}(x) + \sum_i \lambda_{t,i}^{-,n}(x), \quad x \in E_n,$$

is bounded above by a deterministic integrable envelope:

$$\Lambda_t(x) \leq \bar{\Lambda}(t), \quad \int_{(0,T]} \bar{\Lambda}(t) \text{Leb}(dt) < \infty.$$

Then the martingale problem for (39) has a non-explosive cadlag Markov jump-diffusion solution on $[0, T]$.

Proof. This is the standard interlacing construction for a diffusion with a state-dependent jump kernel, combined with thinning of a dominating point process; see, for example, the diffusion existence theorem under local Lipschitz and linear-growth assumptions in [10, 15], the martingale-problem formulation in [7], and interlacing, thinning, and marking constructions in [3, 11, 12]. We give the details because the finite-state-space intuition is misleading here: the state space changes dimension at jumps.

Let

$$m(t) := \int_{(0,t]} \bar{\Lambda}(s) \text{Leb}(ds).$$

Since $m(T) < \infty$, an inhomogeneous Poisson process $\bar{N}_t = N_{m(t)}^0$, obtained by time-changing a unit-rate Poisson process N^0 , has only finitely many candidate times $0 < \tau_1 < \dots < \tau_K \leq T$ almost surely. At times with $\bar{\Lambda}(t) = 0$ define all ratios below to be zero; then domination implies $\Lambda_t(x) = 0$ as well. For $x \in E_n$ define the candidate-transition kernel

$$K_t(x, dy) = \left(1 - \frac{\Lambda_t(x)}{\bar{\Lambda}(t)}\right) \delta_x(dy) + \frac{1}{\bar{\Lambda}(t)} Q_t(dy; x),$$

with the convention $K_t(x, dy) = \delta_x(dy)$ when $\bar{\Lambda}(t) = 0$. Equivalently, a candidate time first passes the thinning test with probability $\Lambda_t(x)/\bar{\Lambda}(t)$ and, conditional on passing, jumps to y according to $Q_t(dy; x)/\Lambda_t(x)$. Because Q_t is the finite atomic kernel in (42), this means choosing one of the finitely many split/delete moves with probability proportional to its rate.

Construct X recursively. Start from $X_0 \in E_n$. On $[0, \tau_1)$ solve the stratum- n SDE

$$dX_t = b_t^n(X_t) dt + \sigma_t^n(X_t) dW_t^n.$$

The local Lipschitz and linear-growth assumptions give a unique non-explosive solution on every compact subinterval. At τ_1 , draw X_{τ_1} from $K_{\tau_1}(X_{\tau_1-}, dy)$. If the mark is $y = X_{\tau_1-}$ there is no real jump; otherwise the new state lies in the adjacent stratum selected by the split/delete atom. Then restart the SDE on the new stratum, using fresh independent Brownian increments, until the next candidate time. Repeating this procedure through τ_K and then solving the last SDE on $[\tau_K, T]$ defines an adapted cadlag process on the whole interval.

Non-explosion follows from the two separate bounds. The jump times cannot accumulate because $\bar{N}_T < \infty$ almost surely. Between those finitely many candidate times, the continuous motion is a finite concatenation of non-explosive SDE solutions. Hence neither the jump component nor the diffusion component can explode before T .

It remains to identify the generator. Let f be a bounded test function whose restriction f_n is C^2 on each stratum, with bounded derivatives on compact time intervals. Ito's formula on each open inter-jump interval gives the continuous compensator

$$\int_{(0,t]} (\mathcal{A}_s f)(X_s) \text{Leb}(ds).$$

For the jump part, the marked candidate process has predictable compensator

$$\bar{\Lambda}(s)K_s(X_{s-}, dy) \text{Leb}(ds).$$

The atom at $y = X_{s-}$ contributes zero to $f(y) - f(X_{s-})$, so the compensator of the actual jump contribution is

$$\int_{(0,t]} \int_{\mathbf{S}_N} [f(y) - f(X_{s-})] Q_s(dy; X_{s-}) \text{Leb}(ds).$$

Since $X_{s-} = X_s$ for Leb-a.e. s , the process

$$\begin{aligned} M_t^f &:= f(X_t) - f(X_0) - \int_{(0,t]} (\mathcal{A}_s f)(X_s) \text{Leb}(ds) \\ &\quad - \int_{(0,t]} \int_{\mathbf{S}_N} [f(y) - f(X_s)] Q_s(dy; X_s) \text{Leb}(ds) \end{aligned}$$

is a martingale. Using (42), the last integral is exactly the two split/delete sums in (39). Therefore X solves the martingale problem for (39). The construction uses only the current state and independent future Brownian, Poisson, and marking randomness after every stopping time, so the resulting solution is Markov by the usual interlacing argument. $\square$

Remark 3.4 (Hazards that blow up near the endpoint). Conditional Branching Flow hazards are often chosen so that all remaining conditional split/deletion events occur by $t = 1$. That can involve non-integrable hazards near the endpoint, but the conditional remaining-event count decreases to zero. For a learned marginal continuous-time sampler, exact theorem statements need either an integrable total-rate envelope, a direct finite-event proof, or a Lyapunov non-explosion proof. This is a genuine mathematical assumption. A safe practical formalization is to steer on $[0, 1 - \varepsilon]$ and handle the last interval by terminal correction or resampling.

Remark 3.5 (Uncapped state space). For $\bigsqcup_{n \geq 1} E_n$, replace the finite cap by a Lyapunov condition: find $V \geq 1$ with compact sublevel sets such that $\mathcal{L}_t V \leq CV$ up to localization. The h-transform theorem still applies, provided the transformed rates also satisfy a non-explosion condition.

4 Question i: When does H solve the backward integro-differential equation?

Let $w > 0$ be a bounded terminal reward weight and define

$$H(t, x) = \mathbb{E}_{t,x}[w(X_T)]. \tag{44}$$

More generally, with reward density ρ with respect to time,

$$H(t, x) = \mathbb{E}_{t,x} \left[\exp \left\{ \mathbf{A}_{t,T}^\rho(X) \right\} w(X_T) \right]. \tag{45}$$

The minimal answer. Under non-explosion and integrability, (44) or (45) is always the right probabilistic object. It is a mild Feynman–Kac solution [19, 20]. The pointwise integro-differential equation is valid when H is regular enough to lie in the domain of the extended generator.

The pointwise equation. If $H_n(t, \cdot)$ is twice continuously differentiable on each stratum, H is differentiable in t , and $\mathcal{L}_t H_t$ is continuous and locally bounded, then H solves

$$\partial_t H(t, x) + \mathcal{L}_t H(t, \cdot)(x) + \rho_t(x) H(t, x) = 0, \quad H(T, x) = w(x). \quad (46)$$

On stratum E_n this is the backward integro-differential system

$$\begin{aligned} 0 = & \partial_t H_n(t, x) + b_t^n(x) \cdot \nabla_n H_n(t, x) + \frac{1}{2} \text{Tr} \left(a_t^n(x) \nabla_n^2 H_n(t, x) \right) + \rho_t^n(x) H_n(t, x) \\ & + \sum_{i=1}^n \lambda_{t,i}^{+,n}(x) [H_{n+1}(t, \text{split}_i x) - H_n(t, x)] \\ & + \sum_{i=1}^n \lambda_{t,i}^{-,n}(x) [H_{n-1}(t, \text{del}_i x) - H_n(t, x)], \quad H_n(T, x) = w_n(x). \end{aligned} \quad (47)$$

Equivalently, in kernel form,

$$0 = \partial_t H(t, x) + \mathcal{A}_t H(t, x) + \rho_t(x) H(t, x) + \int_{\mathbf{S}_N} [H(t, y) - H(t, x)] Q_t(dy; x). \quad (48)$$

Proposition 4.1 (Mild versus classical Feynman–Kac). *Assume the non-explosive Markov jump-diffusion of Lemma 3.3. Let w be bounded measurable and ρ bounded measurable. Then H defined by (45) satisfies the dynamic programming identity*

$$H(s, x) = \mathbb{E}_{s,x} \left[\exp \left\{ \mathbf{A}_{s,t}^\rho(X) \right\} H(t, X_t) \right], \quad s \leq t \leq T. \quad (49)$$

This is the mild Feynman–Kac equation. If, in addition, H has the regularity stated above, then H is a classical solution of (46)–(47).

Proof. The dynamic programming identity is the Markov property plus the tower property:

$$\begin{aligned} H(s, x) &= \mathbb{E}_{s,x} \left[\exp \left\{ \mathbf{A}_{s,T}^\rho(X) \right\} w(X_T) \right] \\ &= \mathbb{E}_{s,x} \left[\exp \left\{ \mathbf{A}_{s,t}^\rho(X) \right\} \mathbb{E} \left[\exp \left\{ \mathbf{A}_{t,T}^\rho(X) \right\} w(X_T) \mid \mathcal{F}_t \right] \right] \\ &= \mathbb{E}_{s,x} \left[\exp \left\{ \mathbf{A}_{s,t}^\rho(X) \right\} H(t, X_t) \right]. \end{aligned}$$

If H is in the generator domain, expand (49) over $t = s + h$ and let $h \downarrow 0$. The order- h terms are $\partial_t H$, $\mathcal{L}_t H$, and ρH , giving (46). Substituting the Branching Flow generator (39) gives (47). $\square$

Remark 4.2 (Useful sufficient regularity). A conservative sufficient condition for a classical equation is: $N < \infty$; each a_t^n is uniformly elliptic on compact sets; coefficients a, b and rates $\lambda^\pm$ are bounded with enough spatial and temporal smoothness; the terminal reward w_n is bounded and sufficiently smooth on each stratum; and compatibility/localization conditions hold at infinity. Without ellipticity or smooth reward, the right object is usually a mild or viscosity solution. The exact sampling theorem below does not require classical PDE regularity; classical regularity is needed only to write the local generator formula pointwise.

5 Question ii: Why do surviving samples have terminal law q^β ?

This is a one-line Bayes argument, but it is worth stating rigorously.

Proposition 5.1 (Terminal survival gives the reward-shifted law). *Let $X_T \sim q$ under the base Branching Flow sampler. Let $w(x) = e^{\beta r(x)}$ and assume $0 < w(x) \leq M < \infty$. After running the base process to T , draw a Bernoulli survival variable S satisfying*

$$\mathbb{P}(S = 1 \mid X_T = x) = \frac{w(x)}{M}.$$

Then, for every measurable $A \subset \mathbb{S}_N$,

$$\mathbb{P}(X_T \in A \mid S = 1) = \frac{\int_A w(x)q(\mathrm{d}x)}{\int_{\mathbb{S}_N} w(x)q(\mathrm{d}x)} = q^\beta(A). \quad (50)$$

Proof. By Bayes' rule,

$$\begin{aligned} \mathbb{P}(X_T \in A \mid S = 1) &= \frac{\mathbb{P}(X_T \in A, S = 1)}{\mathbb{P}(S = 1)} \\ &= \frac{\mathbb{E}[\mathbf{1}_A(X_T)\mathbb{P}(S = 1 \mid X_T)]}{\mathbb{E}[\mathbb{P}(S = 1 \mid X_T)]} \\ &= \frac{\mathbb{E}[\mathbf{1}_A(X_T)w(X_T)/M]}{\mathbb{E}[w(X_T)/M]} \\ &= \frac{\int_A w(x)q(\mathrm{d}x)}{\int w(x)q(\mathrm{d}x)}. \end{aligned}$$

□

This proposition is exact and likelihood-free. It uses only generated samples and reward evaluations. Generalized Doob steering is the Markovian process-level version of the same conditioning: rather than wait until T and reject, it changes the dynamics at time t using the conditional future survival probability $H(t, X_t)$.

6 Question iii: Why is $M_t = H(t, X_t)$ a martingale?

For terminal reward only, set

$$H(t, x) = \mathbb{E}_{t,x}[w(X_T)], \quad M_t = H(t, X_t).$$

The key identity is not merely “ H is a conditional expectation given X_t ”. The rigorous identity is

$$H(t, X_t) = \mathbb{E}[w(X_T) \mid \mathcal{F}_t]. \quad (51)$$

This follows from the Markov property: conditional on the full past $\mathcal{F}_t$, the distribution of the future depends on the past only through X_t .

Then for $s \leq t$,

$$\begin{aligned} \mathbb{E}[M_t \mid \mathcal{F}_s] &= \mathbb{E}[\mathbb{E}[w(X_T) \mid \mathcal{F}_t] \mid \mathcal{F}_s] \\ &= \mathbb{E}[w(X_T) \mid \mathcal{F}_s] \\ &= H(s, X_s) = M_s. \end{aligned} \quad (52)$$

This is the tower property of conditional expectation. Boundedness or integrability of w ensures integrability of M_t .

With reward density ρ with respect to time, the martingale is

$$M_t = \exp \left\{ A_{0,t}^\rho(X) \right\} H(t, X_t), \quad (53)$$

where H is defined by (45). The same tower-property proof gives

$$\begin{aligned} \mathbb{E}[M_t \mid \mathcal{F}_s] &= \exp \left\{ A_{0,s}^\rho(X) \right\} \mathbb{E} \left[\exp \left\{ A_{s,t}^\rho(X) \right\} H(t, X_t) \mid \mathcal{F}_s \right] \\ &= \exp \left\{ A_{0,s}^\rho(X) \right\} H(s, X_s) = M_s. \end{aligned}$$

Remark 6.1 (PDE proof of the same fact). If H is a classical solution of (46), Ito's formula with jumps gives the drift of $H(t, X_t)$ as $(\partial_t + \mathcal{L}_t)H(t, X_t) dt$, which is zero in the terminal-only case. Hence $H(t, X_t)$ is a local martingale; boundedness makes it a true martingale. This is equivalent to the conditional-expectation proof, but the conditional-expectation proof needs fewer regularity assumptions.

7 The exact generalized Doob theorem

Let the base process have initial law μ_0 and terminal law

$$q = \text{Law}_{\mu_0}(X_T).$$

Let $w = e^{\beta r}$ and

$$Z = \mathbb{E}_{\mu_0}[w(X_T)].$$

For a path reward density, set

$$Z_\rho = \mathbb{E}_{\mu_0} \left[\exp \left\{ A_{0,T}^\rho(X) \right\} w(X_T) \right].$$

Theorem 7.1 (Generalized Doob steering for Branching Flows). *Let X be a non-explosive Markov jump-diffusion on S_N with generator (39). Let $w > 0$ be bounded and let ρ be a bounded reward density with respect to Lebesgue time. Define H by (45), assume $H > 0$, and assume the transformed process below is non-explosive. Define a new path measure by*

$$\frac{d\mathbb{P}^H}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = \frac{1}{Z_\rho} \exp \left\{ A_{0,t}^\rho(X) \right\} H(t, X_t). \quad (54)$$

Then $\mathbb{P}^H$ is a Markov law. This path-law statement is the same normalization principle as the generalized Doob transform of a non-conservative semigroup [23] and Léonard's Markov transform $f(X_0)e^{-A_{0,1}^V(X)}g(X_1)R$ [14]. Its initial law is

$$\mu_0^H(dx) = \frac{H(0, x)}{Z_\rho} \mu_0(dx), \quad (55)$$

and its transition kernel is

$$P_{s,t}^H f(x) = \frac{1}{H(s, x)} \mathbb{E}_{s,x} \left[\exp \left\{ A_{s,t}^\rho(X) \right\} H(t, X_t) f(X_t) \right]. \quad (56)$$

The terminal law is

$$\mathbb{P}^H(X_T \in A) = \frac{\mathbb{E}_{\mu_0} \left[\mathbf{1}_A(X_T) \exp \{ A_{0,T}^\rho(X) \} w(X_T) \right]}{Z_\rho}. \quad (57)$$

In particular, if $\rho \equiv 0$, then

$$\mathbb{P}^H(X_T \in A) = q^\beta(A) = \frac{\int_A e^{\beta r(y)} q(dy)}{\int e^{\beta r(y)} q(dy)}. \quad (58)$$

If H is regular enough for the generator calculation, then for time-independent test functions f ,

$$\boxed{\mathcal{L}_t^H f(x) = \frac{1}{H(t, x)} \mathcal{L}_t(H_t f)(x) - f(x) \frac{1}{H(t, x)} \mathcal{L}_t H_t(x)}, \quad (59)$$

where $H_t(x) = H(t, x)$.

Proof. The process

$$M_t = \exp \left\{ A_{0,t}^\rho(X) \right\} H(t, X_t)$$

is a positive martingale by the argument in Section 5. Since $\mathbb{E}[M_0] = Z_\rho$, (54) defines a consistent change of measure. The initial law follows by taking $t = 0$.

For $s < t$, Bayes' rule and the Markov property give

$$\mathbb{E}^H[f(X_t) \mid X_s = x] = \frac{1}{H(s, x)} \mathbb{E}_{s,x} \left[\exp \left\{ A_{s,t}^\rho(X) \right\} H(t, X_t) f(X_t) \right],$$

which is (56). Therefore the transformed law is Markov. At $t = T$, $H(T, x) = w(x)$, so (54) gives (57). Setting $\rho = 0$ gives (58).

For the generator, expand (56) over $[t, t+h]$. Using the classical equation (46),

$$\begin{aligned} \mathcal{L}_t^H f(x) &= \frac{1}{H_t(x)} [(\partial_t + \mathcal{L}_t + \rho_t)(H_t f)(x)] - f(x) \frac{1}{H_t(x)} [(\partial_t + \mathcal{L}_t + \rho_t)H_t(x)] \\ &= \frac{1}{H_t(x)} \mathcal{L}_t(H_t f)(x) - f(x) \frac{1}{H_t(x)} \mathcal{L}_t H_t(x), \end{aligned}$$

where f has no explicit time-dependence. This proves (59). $\square$

Remark 7.2 (The initial law issue). Exact process-level steering generally changes the initial law to (55). If X_0 is deterministic, this correction is invisible. If $X_0 \sim \mu_0$ is random, keeping the original initial law and changing only the generator is generally not exact. One can implement the initial tilt by resampling from estimates of $H(0, x)$, or by using a sequential Monte Carlo/FK resampling scheme; see Del Moral's monograph and the backward-particle interpretation of Feynman–Kac formulae [4, 29].

8 Question iv: Why does the diffusion part follow from the product rule?

We now compute (59) explicitly. Fix a stratum and suppress n, t in the notation. Let

$$\mathcal{A}f = b \cdot \nabla f + \frac{1}{2} a : \nabla^2 f.$$

Then

$$\begin{aligned} \nabla(Hf) &= H \nabla f + f \nabla H, \\ \nabla^2(Hf) &= H \nabla^2 f + f \nabla^2 H + \nabla H \otimes \nabla f + \nabla f \otimes \nabla H. \end{aligned}$$

Since a is symmetric,

$$\frac{1}{2}a : (\nabla H \otimes \nabla f + \nabla f \otimes \nabla H) = (a \nabla H) \cdot \nabla f.$$

Therefore

$$\begin{aligned} \mathcal{A}(Hf) &= b \cdot (H \nabla f + f \nabla H) + \frac{1}{2}a : (H \nabla^2 f + f \nabla^2 H) + (a \nabla H) \cdot \nabla f \\ &= H \mathcal{A}f + f \mathcal{A}H + (a \nabla H) \cdot \nabla f. \end{aligned}$$

Substitute into the generalized Doob generator:

$$\begin{aligned} H^{-1} \mathcal{A}(Hf) - f H^{-1} \mathcal{A}H &= \mathcal{A}f + H^{-1} (a \nabla H) \cdot \nabla f \\ &= (b + a \nabla \log H) \cdot \nabla f + \frac{1}{2}a : \nabla^2 f. \end{aligned}$$

This proves the drift shift.

For one jump atom $x \mapsto \Gamma(x)$ with rate $\lambda(x)$,

$$\begin{aligned} \frac{1}{H(x)} \lambda(x) [H(\Gamma x) f(\Gamma x) - H(x) f(x)] - f(x) \frac{1}{H(x)} \lambda(x) [H(\Gamma x) - H(x)] \\ = \lambda(x) \frac{H(\Gamma x)}{H(x)} [f(\Gamma x) - f(x)]. \end{aligned}$$

Thus each jump rate is multiplied by $H(\Gamma x)/H(x)$.

Corollary 8.1 (Explicit steered Branching Flow generator). *For $x \in E_n$, the exact steered generator is*

$$\begin{aligned} (\mathcal{L}_t^H f)_n(x) &= (b_t^n(x) + a_t^n(x) \nabla_n \log H_n(t, x)) \cdot \nabla_n f_n(x) + \frac{1}{2} \text{Tr} \left(a_t^n(x) \nabla_n^2 f_n(x) \right) \\ &\quad + \sum_{i=1}^n \lambda_{t,i}^{+,n}(x) \frac{H_{n+1}(t, \text{split}_i x)}{H_n(t, x)} [f_{n+1}(\text{split}_i x) - f_n(x)] \\ &\quad + \sum_{i=1}^n \lambda_{t,i}^{-,n}(x) \frac{H_{n-1}(t, \text{del}_i x)}{H_n(t, x)} [f_{n-1}(\text{del}_i x) - f_n(x)]. \end{aligned} \tag{60}$$

For Branching Flow rates (41), this means

$$\lambda_{t,i}^{+,H}(x) = h_{\text{split}}(t) R_{\theta,t,i}(x) \frac{H(t, \text{split}_i x)}{H(t, x)}, \tag{61}$$

$$\lambda_{t,i}^{-,H}(x) = h_{\text{del}}(t) \rho_{\theta,t,i}(x) \frac{H(t, \text{del}_i x)}{H(t, x)}. \tag{62}$$

Remark 8.2 (Isotropic Euclidean case). If $a_t^n = v_t I$, then the continuous coordinate dynamics become

$$dX_t = [b_t(X_t) + v_t \nabla \log H(t, X_t)] dt + \sqrt{v_t} dW_t,$$

with split/delete rates (61)-(62). If $a = 0$, there is no continuous drift steering term; only the initial law and jump rates can be tilted. This is an absolute-continuity support issue, not a failure of the theorem.

9 Approximate desirability functions

The exact result uses the exact H . A learned network $\hat{H}$ generally has a backward residual.

Proposition 9.1 (Residual identity). *Let $\hat{H} > 0$ be regular and define the formal h -transformed generator*

$$\hat{\mathcal{L}}_t f = \hat{H}_t^{-1} \mathcal{L}_t(\hat{H}_t f) - f \hat{H}_t^{-1} \mathcal{L}_t \hat{H}_t.$$

Set

$$\epsilon(t, x) = \frac{(\partial_t + \mathcal{L}_t) \hat{H}(t, x)}{\hat{H}(t, x)}.$$

Assuming non-explosion and well-posedness, the process generated by $\hat{\mathcal{L}}$ and initialized from

$$\hat{\mu}_0(dx) \propto \hat{H}(0, x) \mu_0(dx)$$

has path density with respect to the base law proportional to

$$\hat{H}(T, X_T) \exp \left\{ -\mathbf{A}_{0,T}^\epsilon(X) \right\}. \quad (63)$$

If $\hat{H}(T, x) = w(x)$ and $|\epsilon(t, x)| \leq \delta$, then the approximate terminal law $\hat{q}_T$ satisfies

$$\|\hat{q}_T - q^\beta\|_{\text{TV}} \leq \min\{1, e^{2\delta T} - 1\}. \quad (64)$$

Proof. Apply Ito's formula with jumps to

$$\hat{H}(t, X_t) \exp \left\{ -\mathbf{A}_{0,t}^\epsilon(X) \right\}.$$

The drift cancels by the definition of ϵ , so the product is a local martingale. Normalizing it gives the change of measure whose generator is $\hat{\mathcal{L}}$. If $\hat{H}(T) = w$ and $|\epsilon| \leq \delta$, then the extra path factor lies in $[e^{-\delta T}, e^{\delta T}]$. Relative to q^β , the normalized density ratio is therefore bounded between $e^{-2\delta T}$ and $e^{2\delta T}$, which implies (64). $\square$

10 Practical likelihood-free recipe

The theorem suggests the following algorithmic implementation.

1. Use the base Branching Flow to generate triples (t, X_t, X_T) .
2. Evaluate $w(X_T) = \exp\{\beta r(X_T)\}$.
3. Train a positive network $H_\phi(t, x)$ by rollout regression or temporal-difference/FK objectives so that

$$H_\phi(t, x) \approx \mathbb{E}[w(X_T) \mid X_t = x].$$

4. During sampling, replace

$$\begin{aligned} b_t & \text{ by } b_t + a_t \nabla \log H_\phi(t, x), \\ \lambda_{t,i}^+ & \text{ by } \lambda_{t,i}^+ \frac{H_\phi(t, \text{split}_i x)}{H_\phi(t, x)}, \quad \lambda_{t,i}^- & \text{ by } \lambda_{t,i}^- \frac{H_\phi(t, \text{del}_i x)}{H_\phi(t, x)}. \end{aligned}$$

5. If X_0 is random, include initial resampling proportional to $H_\phi(0, X_0)$, or use FK/SMC resampling across the path.

No step requires evaluating the terminal density $q(x)$ or a path likelihood. The only terminal oracle is the reward. This is the algorithmic counterpart of normalized Feynman–Kac semigroups and driven-process methods used in rare-event simulation, particle methods, and large-deviation steering [4, 29, 23, 30].

11 Compact theorem statement for a paper

Let X be the non-explosive Branching Flow Markov process on $\bigsqcup_{n=1}^N (\mathbb{R}^d)^n$ with generator (39), terminal law q , and initial law μ_0 . For any bounded positive reward weight $w = e^{\beta r}$, set $H(t, x) = \mathbb{E}[w(X_T) \mid X_t = x]$ and $Z = \mathbb{E}[w(X_T)]$. If H is positive and regular, then the generalized-Doob-transformed Branching Flow with initial law $H(0, x)\mu_0(dx)/Z$, Euclidean drift $b + a\nabla \log H$, and split/delete rates multiplied by $H(t, \text{post-jump})/H(t, \text{pre-jump})$ has terminal law $w(x)q(dx)/Z$. The construction is likelihood-free: H is a conditional reward-to-go under the base sampler.

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